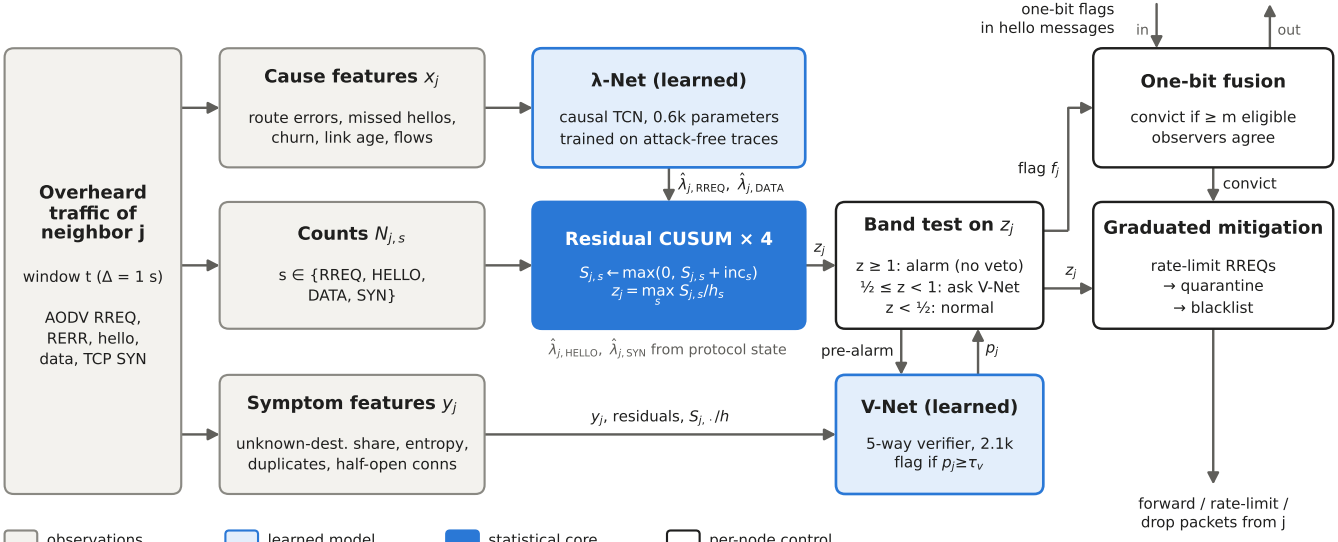
Overheard traffic of neighbor j
window t (Δ = 1 s)
AODV RREQ, RERR, hello, data, TCP SYN
Cause features x_j
route errors, missed hellos, churn, link age, flows
Counts $N_{j,s}$
s ∈ {RREQ, HELLO, DATA, SYN}
Symptom features y_j
unknown-dest. share, entropy, duplicates, half-open conns
λ-Net (learned)
causal TCN, 0.6k parameters trained on attack-free traces
$\hat{\lambda}_{j,\mathrm{RREQ}}$, $\hat{\lambda}_{j,\mathrm{DATA}}$
Residual CUSUM × 4
$S_{j,s} \leftarrow \max(0, S_{j,s} + \mathrm{inc}_s)$
$z_j = \max_s S_{j,s}/h_s$
$\hat{\lambda}_{j,\mathrm{HELLO}}$, $\hat{\lambda}_{j,\mathrm{SYN}}$ from protocol state
z_j
y_j, residuals $S_{j,\cdot}/h$
Band test on z_j
z ≥ 1: alarm (no veto)
½ ≤ z < 1: ask V-Net
z < ½: normal
pre-alarm
p_j
V-Net (learned)
5-way verifier, 2.1k
flag if $p_j \geq \tau_v$
flag f_j
z_j
one-bit flags in hello messages
in
out
One-bit fusion
convict if ≥ m eligible observers agree
convict
Graduated mitigation
rate-limit RREQs
→ quarantine
→ blacklist
forward / rate-limit / drop packets from j
observations
learned model
statistical core
per-node control